



asyncio is Python's built-in library for writing concurrent code using the async/await syntax. It is single-threaded and single-process, using cooperative multitasking to handle I/O-bound tasks without blocking the main execution. [1, 2, 3]

Why Use asyncio?

It dramatically improves the performance of network-heavy or I/O-bound programs (e.g., API requests, web scraping, database calls). Instead of idling while waiting for a response, your program pauses the current task and does other useful work. [1, 2, 3, 4, 5]

Note: For CPU-heavy math computations, consider Python's multiprocessing module instead. [1, 2]

Core Components

1. **Event Loop:** The "orchestrator" that manages and distributes the execution of different tasks.
2. **Coroutines:** Functions defined with async def that can be paused and resumed.
3. **Tasks:** Wrappers used to schedule coroutines concurrently. [1, 2]

Example: Hello World

To execute an asynchronous function, use asyncio.run() to start the event loop: [1, 2]

python

```
import asyncio
```

```
async def main():
    print("Hello...")
    await asyncio.sleep(1) # Yields control to other tasks
    print("...World!")
```

```
asyncio.run(main())
```

Use code with caution.

Running Tasks Concurrently

To run multiple tasks simultaneously, you can use `asyncio.gather()`: [\[1, 2\]](#)

python

```
import asyncio

async def fetch_data(id, delay):
    print(f"Task {id}: Starting")
    await asyncio.sleep(delay)
    print(f"Task {id}: Done")
    return f"Data {id}"

async def main():
    # Runs all three fetch_data functions concurrently
    results = await asyncio.gather(
        fetch_data(1, 2),
        fetch_data(2, 1),
        fetch_data(3, 3)
    )
    print(results)

asyncio.run(main())
Use code with caution.
```

Official Resources

For complete guides, tutorials, and a deeper dive into the APIs, check out these trusted resources:

- [Python asyncio Documentation](#): The official reference for all high-level and low-level methods.
- [Real Python's Async I/O Guide](#): A comprehensive hands-on walkthrough with practical examples. [\[1, 2, 3, 4\]](#)